A UB-ETDA-Based Spaceborne SAR Parameter Design Method for Enhanced Extended Target Detection Performance

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***Abstract*—For extended target detection, most existing spaceborne synthetic aperture radar (SAR) parameter design methods primarily focus on ensuring normal system operation, with limited consideration of the impact of SAR parameters on extended target detection performance. To enhance the extended target detection performance of spaceborne SAR from a parameter design perspective, a novel parameter design method based on an upper bound on extended target detection accuracy (UB-ETDA) is proposed, which comprehensively accounts for key factors affecting extended target detection performance, including aspect-dependent scattering characteristics of the extended target and spatial correlation properties of clutter. By employing UB-ETDA as a new performance metric to guide spaceborne SAR parameter design, the extended target detection performance of the system can be effectively improved. First, a statistical signal model for spaceborne SAR extended target detection is developed, systematically characterizing the stochastic properties of the extended target, background clutter, and noise components in SAR images. Then, the expression of UB-ETDA for SAR system is rigorously derived, along with a stable and statistically efficient numerical computation method. Finally, with UB-ETDA as the primary performance metric and existing performance metrics as constraints, a spaceborne SAR parameter design method is proposed for improving extended target detection performance. Simulation results validate the effectiveness of the proposed method.**

***Index Terms*—Detection accuracy upper bound, extended target detection, parameter design, spaceborne synthetic aperture radar (SAR).**

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I. Introduction

SPACEBORNE synthetic aperture radar (SAR) has become an indispensable tool in target detection due to its all-weather and day-and-night imaging capability. As the resolution of spaceborne SAR systems continues to improve, the detailed scattering characteristics of targets are becoming more clearly observable. Targets that were once approximated as point scatterers now exhibit spatially extended features, making the detection of extended targets increasingly important. However, due to the unique imaging mechanism of SAR, strong background clutter that may obscure the target, together with significant variation in target scattering characteristics under different observation geometries, may limit the detection performance of extended targets [1].

Current research on SAR extended target detection primarily focuses on SAR image processing, employing either traditional methods [2], [3] or deep-learning-based approaches [4], [5]. However, the performance of image-domain SAR target detection fundamentally depends on the quality of the acquired SAR images, specifically the fidelity of extended target scattering characteristics and the effectiveness of clutter suppression. Extensive studies have demonstrated that quality of SAR images imposes fundamental bounds on target detection performance [6], [7], [8], [9]. The SAR image acquisition process is the primary factor shaping the image quality [10], [11], and loss of extended target information or additional clutter interference introduced during image acquisition will lead to irreversible degradation of the quality of SAR images. Therefore, it is necessary to work at the acquisition stage through appropriate SAR parameter design, thereby providing effective support for enhanced extended target detection performance.

Spaceborne SAR parameter design is guided by performance metrics, based on which beam pointing parameters, signal parameters, operating modes, and platform parameters are jointly configured and optimized to meet the performance requirements [12]. A critical step in the design process is the development of performance metrics that are closely aligned with the desired observation performance [11], [13]. For detection missions involving extended targets, metrics that effectively capture determining factors of SAR detection performance and reasonably characterize the detection capability provide more effective guidance for parameter design.

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Fundamental performance metrics employed in existing spaceborne SAR parameter design studies include noise-equivalent sigma zero (NESZ), signal-to-noise ratio (SNR), azimuth and range ambiguity ratios (AASRs and RASR), spatial resolution, and the signal-timing constraints imposed by the zebra map [12], [14]. For small satellite SAR, average operating power, data rate, NESZ, resolution, and instrument mass are used as parameter design performance metrics, and the Pareto optimal parameter designs are identified from a large set of parameter designs [15]. For Geosynchronous SAR, SNR, resolution, and resolution direction angle are used to design the receiver's motion parameters [16]; AASR and the SNR scaling factor are used to design the receiver channel configuration [17]; and NESZ, azimuth resolution, and signal timing relationships are adopted to guide radar parameter design [18]. For the Radar for Earth Observation by synthetic aperture DIstributed on a cluster of cubesats equipped with high-technology micro-propellers for new Operative services (RODiO) mission, system requirements such as NESZ and ambiguity-to-signal ratio are used to derive the required number of receivers and the antenna sizes [19].

Recent studies [20], [21] attempted to incorporate extended target detection application requirements into SAR parameter design, particularly for maritime surveillance. In [20], an application-driven SAR parameter design approach is proposed, which transforms application requirements, such as the minimum detectable ship size, detection probability, and false-alarm probability, into constraints on key SAR performance metrics, including NESZ and resolution. In [20], [21], SAR image is modeled as the superposition of a ship target image and system thermal noise, with clutter effects intentionally ignored; the target image pixels are assumed to be independent and identically distributed, and a specific target detection method is adopted, which consists of a cascade of a pixel-by-pixel detector and 2-D binary integration, to derive explicit expressions relating detection and false-alarm probabilities to NESZ and resolution, allowing direct mapping from application requirements to SAR performance metrics. However, these studies exhibit several limitations. First, noise is treated as the dominant interference, while background clutter, which also significantly affects extended target detection performance [22], is ignored. Second, the modeling of extended targets is simplified, where all target pixels are assumed to be i.i.d., leading to loss of the spatial structural information of the target. As the performance is influenced not only by target intensity but also by its spatial structural information [23], [24], such simplified modeling fails to adequately represent the scattering behavior of practical extended targets. Finally, when establishing the quantitative relationship between application requirements and performance metrics, these studies focus on a particular simple detector; due to the limitation to the form of the detector, it is difficult to generalize the derived quantitative relationship to other types of extended target detection methods.

To address the above issues, an upper bound on extended target detection accuracy (UB-ETDA) for SAR is presented. The UB-ETDA provides a detector-independent quantitative relationship between extended target detection performance and SAR parameters. Leveraging this property, a spaceborne SAR parameter design method is proposed, in which both clutter effects and the spatial structural characteristics of extended targets are explicitly incorporated. The main contributions of this article are as follows.

1) A signal model is established, capable of discriminating extended targets, background clutter, and noise, which are considered as random processes. The diversity of extended target scattering characteristics with respect to target orientation angle is explicitly modeled by representing the target as a stochastic distribution composed of multiaspect scattering signatures. The spatial distribution characteristics of clutter are also taken into account by analyzing its multidimensional joint probability density function (JPDF).
2) A closed-form analytical expression of the UB-ETDA is derived through rigorous theoretical analysis, and a stable, statistically efficient numerical computation method for UB-ETDA is further developed. The UB-ETDA, as a function of SAR parameters, characterizes the achievable detection performance limit under a given observation scenario. Since its derivation does not rely on any specific image-domain detection method, the resulting bound is method-independent and applicable to various image-domain detectors, demonstrating strong generalizability. Moreover, the UB-ETDA explicitly incorporates multiple key factors affecting extended target detection performance, including clutter and noise interference, as well as the extended scattering characteristics of the target and their dependence on the observation geometry. Consequently, it provides a theoretical basis for evaluating and optimizing spaceborne SAR parameters for extended target detection.
3) By integrating the proposed UB-ETDA with existing performance metrics, a constrained parameter optimization problem for spaceborne SAR extended target detection is established. Solving the formulated constrained optimization problem yields optimal parameters that improve the extended target detection accuracy of SAR systems.

The remainder of this article is organized as follows. Section II establishes the signal model for extended target detection. Section III derives the UB-ETDA expression and develops a stable, statistically efficient numerical computation method for UB-ETDA. Section IV presents a parameter design method for spaceborne SAR extended target detection by integrating the UB-ETDA with existing performance metrics, along with a method for obtaining the optimal parameters. Section V provides simulations to validate both the proposed UB-ETDA and the SAR parameter design method. Finally, Section VI concludes this article.

II. Signal Model for SAR Extended Target Detection

In this section, the signal model for SAR extended target detection is constructed. First, the detection process for extended targets in SAR is analyzed, and a corresponding

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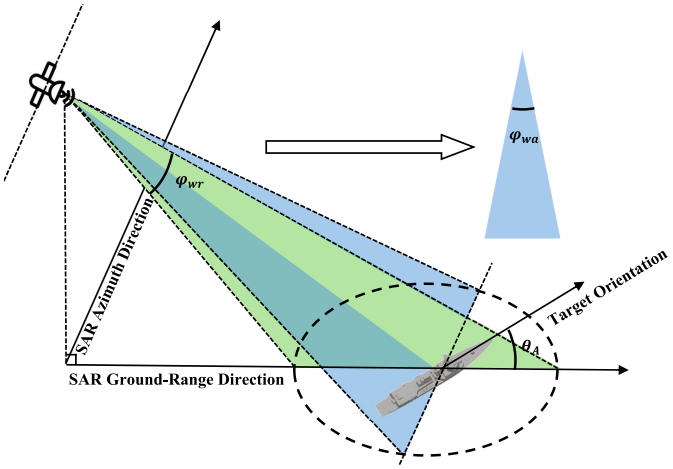


Fig. 1. Observation geometry of an extended target in SAR.

system block diagram is developed. Then, the SAR imaging signal models for the extended target, clutter, and noise components are established. Finally, the statistical distribution of SAR image is derived, providing the mathematical foundation for subsequent analysis of UB-ETDA.

A. SAR Extended Target Detection Process

The observation geometry of an extended target in SAR is illustrated in Fig. 1. The platform motion direction defines the SAR azimuth direction, while the ground direction perpendicular to the azimuth direction is referred to as the ground-range direction. During observation, the SAR antenna illuminates the scene through an electromagnetic (EM) beam, where the beamwidths in the range and azimuth directions are denoted by φ_{wr} and φ_{wa}, respectively, as shown in Fig. 1.

An extended target may exhibit different orientations with respect to the SAR platform. The angle between the target orientation and the ground range direction is denoted by θ_A. Variations in θ_A result in different scattering characteristics of the extended target [25].

The detection process is illustrated by the block diagram shown in Fig. 2.

The SAR system is assumed to operate in the stop-and-go mode, and transmits a waveform and receives the corresponding backscattered signals along the range direction (fast time τ). As the platform moves along the azimuth direction (slow time η), a sequence of backscattered signals is acquired. The backscattered signals are corrupted by the receiver thermal noise $N(\tau,\eta)$, yielding SAR echoes. The impulse response of the acquisition process of backscattered signals, denoted by $h_{\text{Echo}}(\tau,\eta)$, is described in [14] and [22]. The SAR echoes are processed by the imaging processing system $h_{\text{Process}}(\tau,\eta)$ to generate a focused, complex-valued SAR image $I(\tau,\eta)$. The detailed expressions of $h_{\text{Process}}(\tau,\eta)$ can be found in [26]. Accordingly, when the target is present, the SAR image $I_1(\tau,\eta)$ can be decomposed into a target component $I_T(\tau,\eta)$, a clutter component $I_C(\tau,\eta)$, and a noise component $I_N(\tau,\eta)$

$$I_1(\tau,\eta) = I_T(\tau,\eta) + I_C(\tau,\eta) + I_N(\tau,\eta) \tag{1}$$

where

$$\begin{cases} I_T(\tau,\eta) = T(\tau,\eta) * h_{\text{All}}(\tau,\eta) \\ I_C(\tau,\eta) = C(\tau,\eta) * h_{\text{All}}(\tau,\eta) \\ I_N(\tau,\eta) = N(\tau,\eta) * h_{\text{Process}}(\tau,\eta). \end{cases} \tag{2}$$

$T(\tau,\eta)$ and $C(\tau,\eta)$ denote the scattering characteristics of the extended target and the background clutter. $h_{\text{All}}(\tau,\eta)$ denotes the overall impulse response of the SAR image acquisition system

$$h_{\text{All}}(\tau,\eta) = h_{\text{Echo}}(\tau,\eta) * h_{\text{Process}}(\tau,\eta) \tag{3}$$

where * is the linear convolution operation. When target is absent, $I_0(\tau,\eta)$ is composed of $I_C(\tau,\eta)$ and $I_N(\tau,\eta)$ only.

After obtaining the SAR image, the target detector typically scans the image using a sliding window and decides whether a target is present or not based on the pixels within each detection window. Let a binary random variable $V \in \{0,1\}$ denote the true target status ($V = 1$ if present, $V = 0$ if absent), and another binary random variable $\hat{V} \in \{0,1\}$ represent the detector output ($\hat{V} = 1$ if a target is declared, $\hat{V} = 0$ otherwise). Detection is correct when $\hat{V} = V$, and incorrect otherwise.

The accuracy of SAR extended target detection is limited by the image quality, which is determined jointly by the echo formation process $h_{\text{Echo}}(\tau,\eta)$ and the imaging process $h_{\text{Process}}(\tau,\eta)$, which are functions of SAR parameters [14], [22], [26]. Thus, the design of SAR parameters has a critical impact on extended target detection performance.

B. SAR Image Signal Models of the Extended Target, Clutter, and Noise Components

Usually, the three SAR image components $I_T(\tau,\eta)$, $I_C(\tau,\eta)$ and $I_N(\tau,\eta)$ are mutually independent. Therefore, the statistical characteristics of the SAR image can be obtained by separately deriving the statistics of $I_T(\tau,\eta)$, $I_C(\tau,\eta)$ and $I_N(\tau,\eta)$. Since extended target detection jointly exploits multiple SAR image pixels within the detection window, it is necessary to analyze the multidimensional JPDF of N_d samples of $I_T(\tau,\eta)$, $I_C(\tau,\eta)$ and $I_N(\tau,\eta)$, respectively, in order to accurately characterize the signal model for SAR extended target detection.

The JPDF of $I_N(\tau,\eta)$ is analyzed first. Noise $N(\tau,\eta)$ is modeled as a zero-mean stationary Gaussian process [27] with 2-D power spectral density (PSD) $S_N(u,v)$, where u and v are frequency-domain variables corresponding to τ and η. According to the properties of random processes passing through linear time-invariant systems [28], the PSD of $I_N(\tau,\eta)$ can be expressed as

$$S_{I_N}(u,v) = S_N(u,v)\,|H_{\text{Process}}(u,v)|^2 \tag{4}$$

where $H_{Process}(u,v)$ is the 2-D Fourier transform of $h_{\text{Process}}(\tau,\eta)$. Consequently, the noise variance $\sigma^2_{I_N}$ at each time instant of $I_N(\tau,\eta)$ can be expressed as

$$\sigma^2_{I_N} = \int_{-\frac{B_a}{2}}^{\frac{B_a}{2}} \int_{-\frac{B_r}{2}}^{\frac{B_r}{2}} S_{I_N}(u,v)\,\mathrm{d}u\mathrm{d}v \tag{5}$$

where B_r and B_a are the radar signal bandwidths in range and azimuth directions, respectively. $I_N(\tau,\eta)$is typically modeled as a circular complex random process [29], and the probability density function (PDF) of complex-valued variable $I_N(\tau,\eta)$

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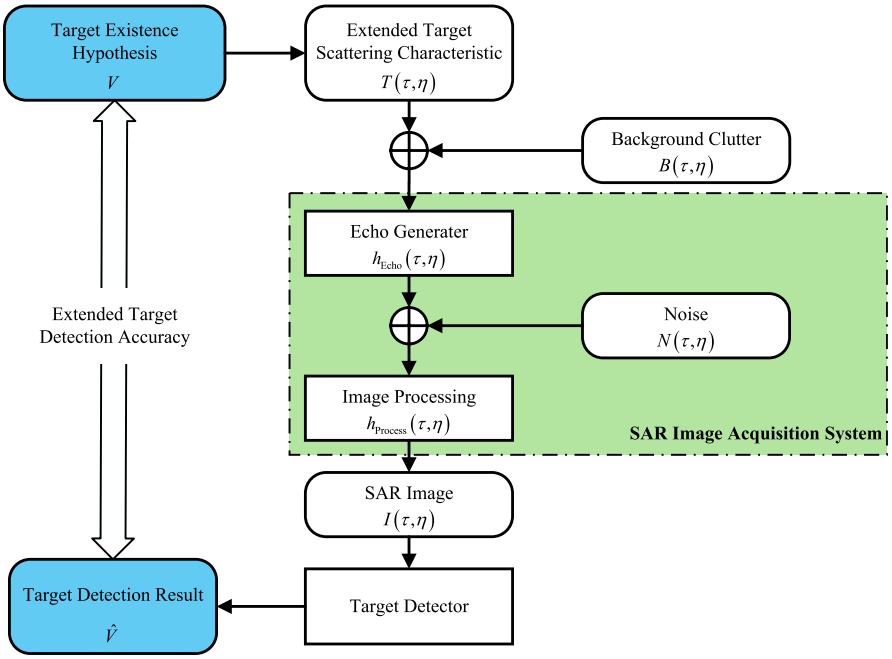


Fig. 2. Block diagram of SAR extended target detection.

at any time instant is $\mathcal{CN}(0,\sigma_{I_N}^2)$. Typically, complex-valued samples at different time instants in $I_N(\tau,\eta)$ can be assumed to be independent and identically distributed [30]. Therefore, the JPDF of any N_d samples of $I_N(\tau,\eta)$ can be expressed as the product of the single instant PDF.

By taking N_d complex-valued samples $I_N(\tau_1,\eta_1)$, $I_N(\tau_2,\eta_2),\ldots,I_N(\tau_{N_d},\eta_{N_d})$ of $I_N(\tau,\eta)$ and separating their real and imaginary parts, a $2N_d$ dimension real-valued random vector $\mathbf{i}_N$ is obtained. $\mathbf{i}_N$ follows a zero-mean multivariate Gaussian distribution. The JPDF of $\mathbf{i}_N$ is given by

$$\mathbf{i}_N \sim \mathcal{N}\left(\mathbf{0},\mathbf{\Sigma}_{I_N}\right) \tag{6}$$

where the covariance matrix $\mathbf{\Sigma}_{I_N}$ is diagonal with identical diagonal elements $(\sigma_{I_N}^2)/2$. In SAR extended target detection, the statistical characteristics of both $I_T(\tau,\eta)$ and $I_C(\tau,\eta)$ are generally complex. Therefore, an SAR extended target signal model that accounts for target orientation and an SAR clutter signal model incorporating spatial correlation are analyzed below.

1) Signal Model of Extended Target Component Considering Target Orientation: The scattering characteristics of a target observed at different θ_A not only reflect changes in its overall orientation, but also capture the spatial distribution of its internal components, as well as the aspect-dependent variations in the backscattering coefficients of individual components. As the orientation of target is generally unknown, the target orientation angle θ_A can be treated as a random variable. Therefore, in the modeling, it is necessary to account for the variations in target scattering characteristics caused by random changes in target orientation.

Existing SAR target modeling approaches often rely on simplified assumptions. Some methods model the target $T(\tau,\eta)$ as a stationary random process [22], [27], ignoring the nonstationary behavior caused by heterogeneous scattering properties of the extended target internal components; others adopt deterministic target templates [6], without accounting for the randomness of target observation aspects in SAR target detection.

To more accurately characterize the stochastic properties of extended targets, $T(\tau,\eta)$ is considered as a random process driven by target orientation. Random variations of target orientation over $[0,2\pi)$ lead to stochastic fluctuations in the value of $T(\tau,\eta)$ [25], [31]. For a given angle θ_A, the 2-D complex scattering characteristic function of the target is denoted by $t(\tau,\eta;\theta_A)$, which is a deterministic template describing the scattering characteristics of the target. For different orientations, the spatial support of the scattering template $t(\tau,\eta;\theta_A)$ may vary. The maximum support lengths of $t(\tau,\eta;\theta_A)$ along the τ and η axes over all possible orientations are denoted as τ_{max} and η_{max}, respectively. To avoid target energy loss caused by template truncation, $T(\tau,\eta)$ is uniformly defined over a spatial region of size $\tau_{max} \times \eta_{max}$. Under a given θ_A, the target component in the SAR image $I_T(\tau,\eta)$ can then be expressed as

$$i_t(\tau,\eta;\theta_A) = t(\tau,\eta;\theta_A) * h_{\text{All}}(\tau,\eta). \tag{7}$$

As an approximation, the continuous variable θ_A can be approximated by a discrete random variable $\hat{\theta}_A$ obtained by uniform quantization. Accordingly, the target scattering characteristics are analyzed at a finite set of discrete target orientation angles. Select N_d complex-valued samples $I_T(\tau_1,\eta_1)$, $I_T(\tau_2,\eta_2),\ldots,I_T(\tau_{N_d},\eta_{N_d})$ of $I_T(\tau,\eta)$ within $\tau_{max} \times \eta_{max}$, and form a $2N_d$-dimensional real-valued random vector $\mathbf{i}_T$ by concatenating the real and imaginary parts of these samples. The probability distribution of $\mathbf{i}_T$ can be obtained as

$$\mathrm{P}\left[\mathbf{i}_T = \mathbf{i}_T\left(\hat{\theta}_A^s\right)\right] = \frac{1}{N_\theta}\,(s = 1,2,\ldots,N_\theta) \tag{8}$$

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where $\hat{\theta}_A^s$ denotes a specific realizations value of the discrete angle $\hat{\theta}_A$, N_θ is the total number of discrete realizations of $\hat{\theta}_A$, and $\mathbf{i}_T(\hat{\theta}_A^s)$ is a $2N_d$-dimensional real-valued vector formed by concatenating the real and imaginary parts of N_d complex-valued samples: $i_t(\tau_1, \eta_1; \hat{\theta}_A^s)$, $i_t(\tau_2, \eta_2; \hat{\theta}_A^s)$, …, $i_t(\tau_{N_d}, \eta_{N_d}; \hat{\theta}_A^s)$. Here, a uniform distribution of the target orientation angle is taken as an example; the method is also applicable to the general nonuniform distributions.

2) Signal Model of Clutter Component Considering the Spatial Correlation: Unlike system noise, clutter in SAR images typically exhibits pronounced spatial correlation [32]. Although the 1-D statistical distribution of SAR clutter has been extensively studied [33], such a 1-D model is insufficient to characterize the spatial correlation structure of clutter. Therefore, it is necessary to analyze the correlation among clutter samples at different time instants when constructing the clutter component signal model. Clutter $C(\tau, \eta)$ typically acts as an interference superimposed on the extended target. To capture its overall spatial texture characteristics, a multidimensional JPDF of the clutter component $I_C(\tau, \eta)$ is employed.

$C(\tau, \eta)$ is modeled as a zero-mean stationary Gaussian process [27]. According to the theory of random processes through linear systems, $I_C(\tau, \eta)$ remains a stationary Gaussian process. For any N_d samples $I_C(\tau_1, \eta_1)$, $I_C(\tau_2, \eta_2)$, …, $I_C(\tau_{N_d}, \eta_{N_d})$ of $I_C(\tau, \eta)$, the JPDF follows a multivariate Gaussian distribution, and therefore can be fully determined from the covariance matrix of these samples.

The 2-D PSD $S_C(u, v)$ of $C(\tau, \eta)$ is employed as prior information. The 2-D PSD of $I_C(\tau, \eta)$ is given by

$$S_{I_C}(u, v) = S_C(u, v) \left| H_{\mathrm{All}}(u, v) \right|^2 \tag{9}$$

where $H_{\mathrm{All}}(u, v)$ is the 2-D Fourier transform of $h_{\mathrm{All}}(\tau, \eta)$. The autocorrelation function $R_{I_C}(\Delta\tau, \Delta\eta)$ of $I_C(\tau, \eta)$ is obtained from the 2-D inverse Fourier transform of $S_{I_C}(u, v)$.

Separate $I_C(\tau, \eta)$ into its real part $\mathrm{Re}[I_C(\tau, \eta)]$ and imaginary part $\mathrm{Im}[I_C(\tau, \eta)]$. Consequently, $R_{I_C}(\Delta\tau, \Delta\eta)$ can be expressed as follows:

$$\begin{aligned} R_{I_C}(\Delta\tau, \Delta\eta) = {} & R_{\mathrm{Re}[I_C]}(\Delta\tau, \Delta\eta) + R_{\mathrm{Im}[I_C]}(\Delta\tau, \Delta\eta) \\ & + j R_{\mathrm{Im}[I_C]\mathrm{Re}[I_C]}(\Delta\tau, \Delta\eta) \\ & - j R_{\mathrm{Re}[I_C]\mathrm{Im}[I_C]}(\Delta\tau, \Delta\eta) \end{aligned} \tag{10}$$

where $R_{\mathrm{Re}[I_C]}(\Delta\tau, \Delta\eta)$ and $R_{\mathrm{Im}[I_C]}(\Delta\tau, \Delta\eta)$ denote the autocorrelation functions of $\mathrm{Re}[I_C(\tau, \eta)]$ and $\mathrm{Im}[I_C(\tau, \eta)]$, respectively, while $R_{\mathrm{Im}[I_C]\mathrm{Re}[I_C]}(\Delta\tau, \Delta\eta)$ and $R_{\mathrm{Re}[I_C]\mathrm{Im}[I_C]}(\Delta\tau, \Delta\eta)$ denote the cross correlation functions between $\mathrm{Re}[I_C(\tau, \eta)]$ and $\mathrm{Im}[I_C(\tau, \eta)]$. Assuming that $I_C(\tau, \eta)$ is a circular random process [28], the autocorrelation functions of its real and imaginary parts and the cross correlation between the real and imaginary parts are given by

$$\begin{aligned} R_{\mathrm{Re}[I_C]}(\Delta\tau, \Delta\eta) &= R_{\mathrm{Im}[I_C]}(\Delta\tau, \Delta\eta) \\ &= \frac{\mathrm{Re}\left[R_{I_C}(\Delta\tau, \Delta\eta)\right]}{2} \end{aligned} \tag{11}$$

and

$$\begin{aligned} R_{\mathrm{Im}[I_C]\mathrm{Re}[I_C]}(\Delta\tau, \Delta\eta) &= -R_{\mathrm{Re}[I_C]\mathrm{Im}[I_C]}(\Delta\tau, \Delta\eta) \\ &= \frac{\mathrm{Im}\left[R_{I_C}(\Delta\tau, \Delta\eta)\right]}{2}. \end{aligned} \tag{12}$$

The vector $\mathbf{i}_C$ consists of the real and imaginary parts of the samples, and its covariance matrix can be partitioned into the following four submatrices:

$$\boldsymbol{\Sigma}_{I_C} = \begin{pmatrix} \boldsymbol{\Sigma}_{\mathrm{Re}[I_C]} & \boldsymbol{\Sigma}_{\mathrm{Re}[I_C]\mathrm{Im}[I_C]} \\ \boldsymbol{\Sigma}_{\mathrm{Im}[I_C]\mathrm{Re}[I_C]} & \boldsymbol{\Sigma}_{\mathrm{Im}[I_C]} \end{pmatrix} \tag{13}$$

where $\boldsymbol{\Sigma}_{\mathrm{Re}[I_C]}$ is the covariance matrix of the real part, $\boldsymbol{\Sigma}_{\mathrm{Im}[I_C]}$ the covariance matrix of the imaginary part, and $\boldsymbol{\Sigma}_{\mathrm{Re}[I_C]\mathrm{Im}[I_C]}$ and $\boldsymbol{\Sigma}_{\mathrm{Im}[I_C]\mathrm{Re}[I_C]}$ represent the cross-covariance matrices between the real and imaginary parts. As both the real and imaginary parts are real-valued, $\boldsymbol{\Sigma}_{\mathrm{Re}[I_C]\mathrm{Im}[I_C]}$ and $\boldsymbol{\Sigma}_{\mathrm{Im}[I_C]\mathrm{Re}[I_C]}$ are transposes of each other. Since $I_C(\tau, \eta)$ is of zero mean, the element in the ith row and jth column of $\boldsymbol{\Sigma}_{\mathrm{Re}[I_C]}$ can be computed as

$$\begin{aligned} \boldsymbol{\Sigma}_{\mathrm{Re}[I_C]}(i, j) &= \mathrm{Cov}\left[\mathrm{Re}\left[I_C(\tau_i, \eta_i)\right] \mathrm{Re}\left[I_C(\tau_j, \eta_j)\right]\right] \\ &= \mathbb{E}\left[\mathrm{Re}\left[I_C(\tau_i, \eta_i)\right] \mathrm{Re}\left[I_C(\tau_j, \eta_j)\right]\right] \\ &= R_{\mathrm{Re}[I_C]}(\tau_i - \tau_j, \eta_i - \eta_j). \end{aligned} \tag{14}$$

The same applies to $\boldsymbol{\Sigma}_{\mathrm{Im}[I_C]}$ and $\boldsymbol{\Sigma}_{\mathrm{Im}[I_C]\mathrm{Re}[I_C]}$

$$\begin{cases} \boldsymbol{\Sigma}_{\mathrm{Im}[I_C]}(i, j) = R_{\mathrm{Im}[I_C]}(\tau_i - \tau_j, \eta_i - \eta_j) \\ \boldsymbol{\Sigma}_{\mathrm{Re}[I_C]\mathrm{Im}[I_C]}(i, j) = R_{\mathrm{Re}[I_C]\mathrm{Im}[I_C]}(\tau_i - \tau_j, \eta_i - \eta_j). \end{cases} \tag{15}$$

So far, the covariance matrix $\boldsymbol{\Sigma}_{I_C}$ has been fully expressed in terms of the autocorrelation function. $\mathbf{i}_C$ follows the following distribution:

$$\mathbf{i}_C \sim \mathcal{N}\left(\mathbf{0}, \boldsymbol{\Sigma}_{I_C}\right). \tag{16}$$

C. Multidimensional JPDF of SAR Image

The JPDF of the SAR image is derived from models of $I_T(\tau, \eta)$, $I_C(\tau, \eta)$ and $I_N(\tau, \eta)$. When the target is absent ($V = 0$), the $2N_d \times 1$ real-valued vector of $I_0(\tau, \eta)$ is given by

$$\mathbf{i}_0 = \mathbf{i}_C + \mathbf{i}_N. \tag{17}$$

Since $\mathbf{i}_C$ and $\mathbf{i}_N$ are mutually independent and both are zero-mean Gaussian, $\mathbf{i}_0$ is also Gaussian. Therefore, its JPDF $f_0(\mathbf{i})$ can be written as

$$f_0(\mathbf{i}) = \mathcal{N}\left(\mathbf{0}, \boldsymbol{\Sigma}_{I_0}\right) \tag{18}$$

with

$$\boldsymbol{\Sigma}_{I_0} = \boldsymbol{\Sigma}_{I_C} + \boldsymbol{\Sigma}_{I_N}. \tag{19}$$

When the target is present ($V = 1$), the real-valued vector of $I_1(\tau, \eta)$ is given by

$$\begin{aligned} \mathbf{i}_1 &= \mathbf{i}_T + \mathbf{i}_C + \mathbf{i}_N \\ &= \mathbf{i}_T + \mathbf{i}_0. \end{aligned} \tag{20}$$

According to the law of total probability, the cumulative distribution function of $\mathbf{i}_1$ is given by

$$\begin{aligned} \mathrm{P}\left[\mathbf{i}_1 \le \mathbf{i}\right] &= \mathrm{P}\left[\mathbf{i}_T + \mathbf{i}_0 \le \mathbf{i}\right] \\ &= \sum_{s=1}^{N_\theta} \mathrm{P}\left[\mathbf{i}_T = \mathbf{i}_T\left(\hat{\theta}_A^s\right)\right] \mathrm{P}\left[\mathbf{i}_0 \le \mathbf{i} - \mathbf{i}_T\left(\hat{\theta}_A^s\right)\right] \\ &= \frac{1}{N_\theta} \sum_{s=1}^{N_\theta} \mathrm{P}\left[\mathbf{i}_0 \le \mathbf{i} - \mathbf{i}_T\left(\hat{\theta}_A^s\right)\right]. \end{aligned} \tag{21}$$

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By differentiating the cumulative distribution function $\mathrm{P}[\mathbf{i}_1 \le \mathbf{i}]$ with respect to vector $\mathbf{i}$, the JPDF of $\mathbf{i}_1$ can be obtained

$$f_1(\mathbf{i}) = \frac{\mathrm{d}}{\mathrm{d}\mathbf{i}} \left\{ \frac{1}{N_\theta} \sum_{s=1}^{N_\theta} \mathrm{P}\left[\mathbf{i}_0 \le \mathbf{i} - \mathbf{i}_T\left(\hat{\theta}_A^s\right)\right] \right\} = \frac{1}{N_\theta} \sum_{s=1}^{N_\theta} \frac{\mathrm{d}}{\mathrm{d}\mathbf{i}} \mathrm{P}\left[\mathbf{i}_0 \le \mathbf{i} - \mathbf{i}_T\left(\hat{\theta}_A^s\right)\right]. \tag{22}$$

Given that $\mathbf{i}_0$ is the zero-mean Gaussian, differentiating $\mathrm{P}[\mathbf{i}_0 \le \mathbf{i} - \mathbf{i}_T(\hat{\theta}_A^s)]$ yields the JPDF $f_0[\mathbf{i} - \mathbf{i}_T(\hat{\theta}_A^s)]$. Therefore, the JPDF of $\mathbf{i}_1$ is in the form of a N_θ-component Gaussian mixture distribution

$$f_1(\mathbf{i}) = \frac{1}{N_\theta} \sum_{s=1}^{N_\theta} \mathcal{N}\left(\mathbf{i}_T\left(\hat{\theta}_A^s\right), \mathbf{\Sigma}_{I_0}\right). \tag{23}$$

III. Upper Bound of SAR Extended Target Detection Accuracy

To guide SAR parameter design toward improving extended target detection performance, it is necessary to develop a performance metric that establishes a quantitative relationship between SAR extended target detection performance and SAR parameters. Moreover, to ensure generality across different target detection methods, SAR parameter design should be conducted without assuming a specific target detector; therefore, such a performance metric should be detector independent.

Based on the statistical distributions of SAR images under target-present and target-absent hypotheses, the extended target detection accuracy (ETDA) can be evaluated. However, ETDA generally depends on a specific detection method and therefore, the UB-ETDA is proposed here as an SAR performance metric, which describes a detector-independent quantitative relationship between extended target detection performance and SAR parameters, by comprehensively accounting for multiple factors affecting the performance.

In the following, first, the mathematical expression of the UB-ETDA is rigorously derived. Then, a stable and statistically efficient numerical method for computing the UB-ETDA is developed and validated through numerical experiments.

A. Derivation of UB-ETDA

Extended target detection determines the presence of a target based on the SAR image pixels contained in the detection window. Therefore, the separability between the JPDF of the SAR image pixels under the target-present and target-absent hypotheses, $f_1(\mathbf{i})$ and $f_0(\mathbf{i})$, respectively, determines the achievable detection accuracy. In the following, the ETDA of SAR is first derived, and an upper bound on the ETDA is then obtained. Finally, the resulting UB-ETDA is further simplified for easier numerical evaluation.

Extended target detection aims to determine the presence of a target by exploiting a $2N_d \times 1$ image sample vector $\mathbf{i}$ constructed from N_d complex-valued SAR image pixels within the detection window. From a mathematical perspective, the detection decision can be formulated as constructing a target existence decision set $\mathcal{A}$ in the observation space $\Omega = \mathbb{R}^{2N_d}$, if the image sample vector $\mathbf{i} \in \mathcal{A}$, the presence of a target is declared; otherwise, the absence of a target is declared. Following directly from the law of total probability, the ETDA can be expressed as:

$$\begin{aligned} \mathrm{ETDA} &= \mathrm{P}\left(\hat{V} = V\right) \\ &= \mathrm{P}\left(\hat{V} = V \mid V = 1\right)\mathrm{P}(V = 1) \\ &\quad + \mathrm{P}\left(\hat{V} = V \mid V = 0\right)\mathrm{P}(V = 0) \end{aligned} \tag{24}$$

where $\mathrm{P}(\hat{V} = V|V = 1)$ and $\mathrm{P}(\hat{V} = V|V = 0)$ denote the probability of a correct decision ($\hat{V} = V$) under the condition $V = 1$ and $V = 0$, respectively. $\mathrm{P}(\hat{V} = V|V = 1)$ and $\mathrm{P}(\hat{V} = V|V = 0)$ can be expressed in integral form

$$\mathrm{ETDA} = p \int_{\mathcal{A}} f_1(\mathbf{i})\, d\mathbf{i} + q \int_{\mathcal{A}^c} f_0(\mathbf{i})\, d\mathbf{i} \tag{25}$$

where p and q denote the probability of $V = 1$ and $V = 0$, respectively, $\int_{\mathcal{A}}(\cdot)\, d\mathbf{i}$ represents integration over the set $\mathcal{A}$, and $\mathcal{A}^C$ denotes the complement of $\mathcal{A}$ in the entire space Ω. It follows further that:

$$\begin{aligned} \mathrm{ETDA} &= p \int_{\mathcal{A}} f_1(\mathbf{i})\, d\mathbf{i} + q\left[1 - \int_{\mathcal{A}} f_0(\mathbf{i})\, d\mathbf{i}\right] \\ &= q + \int_{\mathcal{A}} \left[p f_1(\mathbf{i}) - q f_0(\mathbf{i})\right] d\mathbf{i}. \end{aligned} \tag{26}$$

The value of ETDA depends on the choice of the target existence decision set $\mathcal{A}$. Since the integrand $pf_1(\mathbf{i}) - qf_0(\mathbf{i})$ in (26) takes both positive and negative values, to maximize ETDA, $\mathcal{A}$ is therefore selected as the set over which the integrand is positive. Consequently, the upper bound for ETDA can be written as

$$\mathrm{ETDA} \le q + \int_{\mathcal{A}_*} \left[p f_1(\mathbf{i}) - q f_0(\mathbf{i})\right] d\mathbf{i} \tag{27}$$

where $\mathcal{A}_*$ is the optimal target existence decision set that attains the ETDA upper bound

$$\mathcal{A}_* = \left\{ \mathbf{i} \;\middle|\; p f_1(\mathbf{i}) - q f_0(\mathbf{i}) \ge 0 \right\}. \tag{28}$$

Therefore, the mathematical expression of UB-ETDA can be written as

$$\mathrm{UB-ETDA} = q + \int_{\mathcal{A}_*} \left[p f_1(\mathbf{i}) - q f_0(\mathbf{i})\right] d\mathbf{i}. \tag{29}$$

However, the integration region in (29) does not have an explicit expression, and the computation is relatively complex; therefore, (29) is further simplified. By changing the domain of the integration variables in the integral term of (29), the following expression is obtained

$$\begin{aligned} &\int_{\mathcal{A}_*} \left[p f_1(\mathbf{i}) - q f_0(\mathbf{i})\right] d\mathbf{i} \\ &= \int_{\mathcal{A}_*} p f_1(\mathbf{i})\, d\mathbf{i} - \int_{\mathcal{A}_*} q f_0(\mathbf{i})\, d\mathbf{i} \\ &= p\left[1 - \int_{\mathcal{A}_*^C} f_1(\mathbf{i})\, d\mathbf{i}\right] - q\left[1 - \int_{\mathcal{A}_*^C} f_0(\mathbf{i})\, d\mathbf{i}\right] \\ &= p - q - \int_{\mathcal{A}_*^C} \left[p f_1(\mathbf{i}) - q f_0(\mathbf{i})\right] d\mathbf{i}. \end{aligned} \tag{30}$$

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Substituting (30) into (29)

$$\mathrm{UB-ETDA} = q + \frac{1}{2}\int_{\mathcal{A}_*}\left[pf_1(\mathbf{i}) - qf_0(\mathbf{i})\right] d\mathbf{i} + \frac{1}{2}\left[p - q - \int_{\mathcal{A}_*^C}\left[pf_1(\mathbf{i}) - qf_0(\mathbf{i})\right] d\mathbf{i}\right]. \quad (31)$$

Based on the definition of $\mathcal{A}_*$, it follows that:

$$\int_{\mathcal{A}_*}\left[pf_1(\mathbf{i}) - qf_0(\mathbf{i})\right] d\mathbf{i} = \int_{\mathcal{A}_*}\left|pf_1(\mathbf{i}) - qf_0(\mathbf{i})\right| d\mathbf{i}. \quad (32)$$

The complement of $\mathcal{A}_*$ is defined as

$$\mathcal{A}_*^C = \left\{\mathbf{i} \;\middle|\; pf_1(\mathbf{i}) - qf_0(\mathbf{i}) < 0\right\}. \quad (33)$$

It then follows that:

$$\int_{\mathcal{A}_*^C}\left[pf_1(\mathbf{i}) - qf_0(\mathbf{i})\right] d\mathbf{i} = -\int_{\mathcal{A}_*^C}\left|pf_1(\mathbf{i}) - qf_0(\mathbf{i})\right| d\mathbf{i}. \quad (34)$$

Substituting (32) and (34) into (31) yields

$$\begin{aligned}\mathrm{UB-ETDA} &= \frac{1}{2} + \frac{1}{2}\int_{\mathcal{A}_*}\left|pf_1(\mathbf{i}) - qf_0(\mathbf{i})\right| d\mathbf{i} + \frac{1}{2}\int_{\mathcal{A}_*^C}\left|pf_1(\mathbf{i}) - qf_0(\mathbf{i})\right| d\mathbf{i} \\ &= \frac{1}{2} + \frac{1}{2}\int_{\Omega}\left|pf_1(\mathbf{i}) - qf_0(\mathbf{i})\right| d\mathbf{i}. \end{aligned} \quad (35)$$

Equation (35) presents the computationally convenient expression of UB-ETDA.

As shown by the derivation, UB-ETDA is independent of the specific detection method used and reflects the intrinsic performance limit of an SAR system, making it suitable as a performance metric for spaceborne SAR parameter design.

Although the integrals in (35) are given in an explicit form, the presence of a high-dimensional integral $\int_\Omega |pf_1(\mathbf{i}) - qf_0(\mathbf{i})|\, d\mathbf{i}$ still makes direct numerical evaluation challenging. Section III-B presents an efficient and stable method for numerically evaluating (35).

B. Stable and Statistically Efficient Numerical Method for UB-ETDA

In most extended target detection scenarios the image sample dimension N_d within the detection window is typically larger than 10, so the numerical computation of UB-ETDA inevitably involves high-dimensional integral $\int_\Omega |pf_1(\mathbf{i}) - q\ f_0(\mathbf{i})|\, d\mathbf{i}$ with dimensionality exceeding $2N_d \geq 20$. As a result, conventional grid-based summation approach becomes computationally prohibitive in both time and memory [34].

Therefore, Monte Carlo integration is employed as a statistical numerical integration approach to calculate UB-ETDA. Furthermore, several dedicated strategies are introduced to address the challenges of this high-dimensional integration problem (35), aiming to improve both numerical stability and statistical efficiency of the computation.

First, the integrand of UB-ETDA $|pf_1(\mathbf{i}) - qf_0(\mathbf{i})|$ is composed of $f_0(\mathbf{i})$ and $f_1(\mathbf{i})$, where $f_0(\mathbf{i})$ follows a multivariate Gaussian distribution and $f_1(\mathbf{i})$ follows a Gaussian mixture distribution. Both distributions exhibit highly concentrated probability mass characteristics. Under such conditions, directly using uniform sampling, or sampling distributions mismatched to the main probability mass regions of the integrand, would severely degrade the sampling efficiency [35]. Consequently, for a fixed number of Monte Carlo samples, this results in high estimator variance and pronounced numerical fluctuations, impairing convergence of the integration. Thus, importance sampling is employed by constructing importance distributions that better match the spatial distribution of the integrand [36], [37].

In addition, high-dimensional integrals often involve pronounced scale disparities and statistical correlations across different dimensions, which further exacerbates numerical instability and computational complexity. To mitigate these effects, a whitening transformation is applied prior to integration to eliminate interdimensional correlations and normalize the numerical scales, thereby improving conditioning of the integration space and enhancing numerical stability.

Finally, due to the presence of high-dimensional Gaussian functions in the integrand, exponential operations are inevitably involved. To prevent numerical overflow or underflow of intermediate values, the log-sum-exp stabilization technique is employed.

The detailed implementation of these three steps is presented in the following.

Step 1 (Image Sample Vector Matrix Whitening): A linear whitening transformation $\mathbf{W}$ is applied to decorrelate the image sample vector and rescale its components to unit variance, yielding an identity covariance matrix $\mathbf{I}$. Then, a change of variables is applied to (35)

$$\mathbf{i} = \mathbf{W}^{-1}\mathbf{l} \quad (36)$$

resulting in

$$\mathrm{UB-ETDA} = \frac{1}{2} + \frac{1}{2}\int_{\Omega}\left|pf_1'(\mathbf{l}) - qf_0'(\mathbf{l})\right| d\mathbf{l} \quad (37)$$

where

$$\begin{cases} f_1'(\mathbf{l}) &= \frac{1}{N_\theta}\sum_{s=1}^{N_\theta}\mathcal{N}\left(\mathbf{i}_T'\left(\hat{\theta}_A^s\right), \mathbf{I}\right) \\ f_0'(\mathbf{l}) &= \mathcal{N}(\mathbf{0}, \mathbf{I}) \end{cases} \quad (38)$$

with $\mathbf{i}_T'(\hat{\theta}_A^s)$ denoting $\mathbf{W}\,\mathbf{i}_T(\hat{\theta}_A^s)$.

Step 2 (Selection of the Importance Distribution): To achieve higher statistical efficiency for a fixed sample size, $g(\mathbf{l})$ is selected as the importance distribution

$$g(\mathbf{l}) = q\mathcal{N}(\mathbf{0}, \mathbf{I}) + \frac{p}{N_\theta}\sum_{s=1}^{N_\theta}\mathcal{N}\left(\mathbf{i}_T'\left(\hat{\theta}_A^s\right), \mathbf{I}\right). \quad (39)$$

According to the importance sampling technique, (38) can be written as

$$\begin{aligned}\mathrm{UB-ETDA} &= \frac{1}{2} + \frac{1}{2}\int_{\Omega}\frac{\left|pf_1'(\mathbf{l}) - qf_0'(\mathbf{l})\right|}{g(\mathbf{l})}\, g(\mathbf{l})\, d\mathbf{l} \\ &= \frac{1}{2} + \frac{1}{2}\mathrm{E}_{g(\mathbf{l})}\left[\frac{\left|pf_1'(\mathbf{l}) - qf_0'(\mathbf{l})\right|}{g(\mathbf{l})}\right]. \end{aligned} \quad (40)$$

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By generating N_{sample} random samples $(\mathbf{l}_1, \mathbf{l}_2, \ldots, \mathbf{l}_{N_{\text{sample}}})$ from the distribution $g(\mathbf{l})$, the numerical expression for UB-ETDA can be given as

$$\text{UB}-\text{ETDA} = \frac{1}{2} + \frac{1}{2N_{\text{sample}}} \sum_{k=1}^{N_{\text{sample}}} \frac{\left| pf_1'(\mathbf{l}_k) - qf_0'(\mathbf{l}_k) \right|}{g(\mathbf{l}_k)}. \quad (41)$$

The importance distribution $g(\mathbf{l})$ meets two key requirements: it is a Gaussian mixture distribution, making sample generation straightforward; when the target intensity is greater than the clutter intensity (which is generally the case), the variation trend of $g(\mathbf{l})$ closely matches that of the integrand $|pf_1'(\mathbf{l}) - qf_0'(\mathbf{l})|$ over the majority of the domain.

Algorithm 1 Log-Sum-Exp Stabilization Technique

Input: sample vector $\mathbf{l}_k$,
template means $\mathbf{i}_T'(\hat{\theta}_A^s)$, $s = 1, \ldots, N_\theta$

Output: $R(\mathbf{l}_k) = \dfrac{\left| pf_1'(\mathbf{l}_k) - qf_0'(\mathbf{l}_k) \right|}{g(\mathbf{l}_k)}$

1 **Exponent construction:**

$$\mathbf{z} = \left[-\tfrac{1}{2} \left\| \mathbf{l}_k - \mathbf{i}_T'(\hat{\theta}_A^1) \right\|^2, \ldots, -\tfrac{1}{2} \left\| \mathbf{l}_k - \mathbf{i}_T'(\hat{\theta}_A^{N_\theta}) \right\|^2, -\tfrac{1}{2} \|\mathbf{l}_k\|^2 \right]$$

2 **Max-shift normalization:**

$$m(\mathbf{l}_k) = \max(\mathbf{z})$$

3 **Numerator weights:**

$$\mathbf{W}_A = [\, p/N_\theta, \ldots, p/N_\theta, -q \,]^{\mathrm{T}}$$

4 **Denominator weights:**

$$\mathbf{W}_B = [\, p/N_\theta, \ldots, p/N_\theta, +q \,]^{\mathrm{T}}$$

5 **Weighted sum numerator:**

$$A(\mathbf{l}_k) = \left| \exp[\mathbf{z} - m(\mathbf{l}_k)]\, \mathbf{W}_A \right|$$

6 **Weighted sum denominator:**

$$B(\mathbf{l}_k) = \left| \exp[\mathbf{z} - m(\mathbf{l}_k)]\, \mathbf{W}_B \right|$$

7 **return** $R(\mathbf{l}_k) = A(\mathbf{l}_k)/B(\mathbf{l}_k)$

Step 3 (Avoiding Intermediate Overflow or Underflow): For each sample $\mathbf{l}_k$, the ratio

$$R(\mathbf{l}_k) = \frac{\left| pf_1'(\mathbf{l}_k) - qf_0'(\mathbf{l}_k) \right|}{g(\mathbf{l}_k)} \quad (42)$$

must be computed. However, the exponential term of the Gaussian distribution in the integrand may easily cause overflow or underflow during intermediate calculations. To avoid such problem, the log-sum-exp stabilization technique, as illustrated in Algorithm 1, is employed.

The stability and effectiveness of the proposed Monte Carlo integration method are verified through numerical experiments. The Monte Carlo integration method without the proposed numerical improvements is used for comparison, where Step 1 and Step 3 operations are removed, and a random distribution with zero mean and covariance matrix $\mathbf{\Sigma}_{I_0}$ is used to replace the importance sampling distribution in the proposed method.

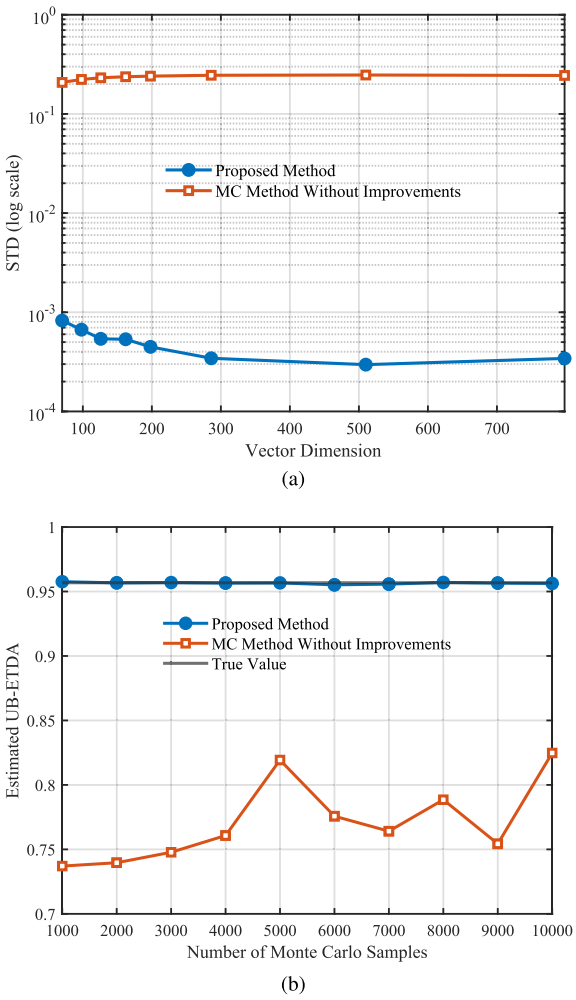


Fig. 3. Statistical efficiency and numerical stability of the proposed Monte Carlo integration method. (a) STD versus image sample vector dimension. (b) Numerical estimation versus sample size under high-dimensional setting.

With the number of samples N_{sample} set to 10 000, each method is repeated 100 times. Fig. 3(a) illustrates the standard deviation (STD) of numerical estimation of the proposed method and the MC method without improvements under different image sample vector dimensions. By comparing the STD of different numerical methods in Fig. 3(a) it can be observed that the proposed method consistently achieves a smaller STD across different image sample vector dimensions, indicating superior statistical efficiency. Moreover, as the vector dimension increases, the estimation accuracy exhibits no noticeable degradation, demonstrating good adaptability to high-dimensional data.

To analyze the numerical stability of the proposed method, the image sample vector dimension is set to 70, and N_{sample} is varied from 1000 to 10 000. The results in Fig. 3(b) indicate that the MC method without improvements consistently exhibits significant deviation from the true value. This is due to the extremely small exponential terms in the probability density in the high-dimensional space, which leads to severe numerical underflow. Moreover, due to the lack of effective numerical protection, the results of the MC method without improvements exhibit substantial fluctuations even with an

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TABLE I
SPACEBORNE SAR PARAMETERS

Category	Parameter	Value / Range
Predefined Parameters	Orbit Altitude H	600 km
	Velocity V	7.55 km/s
	Antenna Depression Angle φ_d	0.61 rad
	Range Beamwidth φ_{wr}	7.6×10^{-3} rad
	Center Frequency f_c	10 GHz
	Transmit Power P_t	2 kW
	Pulse Duration τ	20 μs
	System Loss L_s	6 dB
Design Parameters	Azimuth Beamwidth φ_{wa}	$[5, 15] \times 10^{-3}$ rad
	Chirp Rate γ	$[2.79, 16.74]$ MHz/μs
	PRF f_r	$[2000, 6500]$ Hz

increase in sample size and frequently encounter extreme values, showing no signs of convergence. In contrast, the proposed method closely matches the true value with minimal fluctuations, effectively validating the efficiency of whitening preprocessing and LSE techniques in ensuring the stability of numerical computations.

IV. SPACEBORNE SAR PARAMETER DESIGN METHOD BASED ON UB-ETDA

In this section, a spaceborne SAR parameter design method is proposed, in which UB-ETDA serves as the primary performance metric and is adopted as the objective function, while traditional performance metrics, including resolution, NESZ, RASR, AASR, timing constraints derived from the zebra diagram, and the range observation swath $\mathrm{W}_{\mathrm{obs}}$, are incorporated as constraints to ensure proper system operation.

The parameters of spaceborne SAR are listed in the second column of Table I. These parameters are embedded in the system impulse response $h_{\mathrm{Echo}}(\tau, \eta)$ and $h_{\mathrm{Process}}(\tau, \eta)$, and therefore influence the SAR performance. The UB-ETDA systematically evaluates the impact of these parameters on extended target detection performance.

However, in practical parameter design, not all parameters are treated as free variables. Some parameters are primarily constrained by platform resources and mission planning [12]. For instance, orbit altitude and velocity are determined by revisit time, coverage, and orbital mechanics. The antenna depression angle and range beamwidth are derived from observation planning. The center frequency is selected based on observation scenarios and hardware constraints. In addition, some parameters exhibit relatively monotonic effects on SAR performance, such as transmit power, signal duration, and system loss. Therefore, the above parameters are treated as predefined parameters and assigned typical values.

The azimuth beamwidth, chirp rate, and pulse repetition frequency (PRF) exhibit more complex relationships with extended target detection performance. Specifically, increases in the azimuth beamwidth and chirp rate lead to an enlarged signal bandwidth in the range and azimuth dimensions, respectively, thereby improving the achievable resolution. However, the enlarged bandwidth also introduces higher clutter and noise power. The selection of the PRF must balance the suppression of range and azimuth ambiguities and is further constrained by the zebra map. Accordingly, these three parameters are selected as design parameters. Furthermore, considering that parameter selection is constrained by performance metrics and technological limitations, selectable ranges are imposed on these design parameters.

Consequently, a constrained optimization problem for SAR parameter design is formulated as follows:

$$\begin{aligned} \max_{\mathbf{x}_{\mathrm{par}}} \quad & \mathrm{UB-ETDA}\left(\mathbf{x}_{\mathrm{par}}\right) \\ \text{s.t. } & \mathrm{res}_r = \mathrm{res}_a \le \mathrm{res}_{\mathrm{bound}} \\ & \mathrm{NESZ} \le \mathrm{NESZ}_{\mathrm{bound}} \\ & \mathrm{AASR} \le \mathrm{AASR}_{\mathrm{bound}} \\ & \mathrm{RASR} \le \mathrm{RASR}_{\mathrm{bound}} \\ & f_{\mathrm{zebra}}\left(\mathbf{x}_{\mathrm{par}}\right) = 0 \\ & \mathrm{W}_{\mathrm{obs}} \ge \mathrm{W}_{\mathrm{bound}} \end{aligned} \tag{43}$$

where $\mathbf{x}_{\mathrm{par}}$ is the vector composed of the design parameters (e.g., azimuth beamwidth, chirp rate, and PRF), $\mathrm{res}_{\mathrm{bound}}$, $\mathrm{NESZ}_{\mathrm{bound}}$, $\mathrm{AASR}_{\mathrm{bound}}$, $\mathrm{RASR}_{\mathrm{bound}}$, and $\mathrm{W}_{\mathrm{bound}}$ represent the feasible boundaries of traditional performance metrics, and $f_{\mathrm{zebra}}(\mathbf{x}_{\mathrm{par}})$ denotes the duration of the echo signal being obscured by the nadir point echo and the transmitted signal.

Due to the difficulty of differentiating the objective function UB-ETDA in the optimization problem and the presence of multiple constraints, the ε-differential evolution (ε-DE) algorithm [38] is chosen for solving the problem. Differential Evolution is an intelligent optimization method that explores the solution space by performing mutation, crossover, and selection operations on the population individuals.

To ensure that the solution satisfies the constraints and also avoid missing a solution due to overemphasis on the constraints, the ε-DE algorithm employs the ε- criterion during fitness comparison between the candidate and the original solutions. During the iteration process, when a candidate solution slightly violates the constraints (with the violation within the ε tolerance limit), it is retained if its objective function value is better. As the iteration progresses, the algorithm gradually tightens the tolerance for constraint violations, ultimately converging to the optimal solution that satisfies the constraints. The pseudocode for the ε-DE algorithm can be found in [38].

V. SIMULATION RESULTS

This section first validates the correctness and feasibility of UB-ETDA through simulations. Then, experiments are conducted based on the proposed SAR parameter design method to evaluate its effectiveness in practical scenarios.

A. Simulation Verification of UB-ETDA

In the simulation, the spaceborne SAR operates in a side-looking stripmap mode, and the value of key parameters are listed in the third column of Table I. The azimuth beamwidth, chirp rate, and PRF are selected as design variables, while the remaining predefined parameters are assigned representative values. The spaceborne SAR is designed to achieve a resolution ranging from 0.5 to 3 m. Based on the definitions

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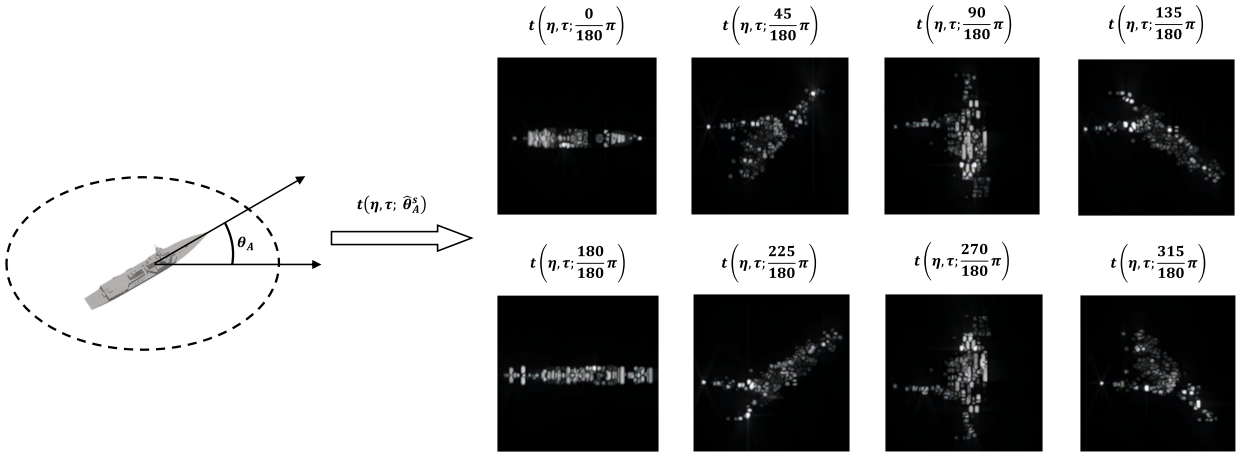


Fig. 4. Backscattering coefficient amplitude images of the ship target at different target orientation angles.

of azimuth and range resolutions, the azimuth beamwidth is constrained within 0.005–0.015 rad, and the chirp rate within 2.79–16.74 MHz/μs. The PRF range is set to 2000 – 6500 Hz, which is consistent with TerraSAR-X [39].

A specific ship target is selected, with a length of 50 m and a width of 8 m. Its EM scattering characteristics are obtained through EM simulation based on its 3-D model. Specifically, the physical optics (PO) method is employed to compute the scattering characteristics at different target orientation angles. The EM simulation is carried out at a center frequency of 10 GHz, with a spatial resolution of 0.2 m and a depression angle of 0.61 rad. Target orientation angle θ_A is uniformly sampled over $[0, 2\pi)$ with an interval of $5/180\pi$ rad, yielding the discretized variable $\hat{\theta}_A$ with $N_\theta = 72$ realizations. Fig. 4 shows the backscattering coefficient amplitude images of the ship target at different $\hat{\theta}_A$. In the simulation, the average backscattering coefficient of the target is set to −17 dB.

The PSD of $C(\eta,\tau)$ is chosen in the form of a typical Lorentzian function [40], [41]

$$S_C(u,v) = \frac{K_c}{\left(1+au^2\right)\left(1+bv^2\right)} \begin{cases} -\dfrac{B_r}{2} \le u \le \dfrac{B_r}{2} \\ -\dfrac{B_a}{2} \le v \le \dfrac{B_a}{2} \end{cases} \tag{44}$$

where a and b denote the correlation lengths in the range and azimuth directions, respectively, and B_r and B_a represent the radar signal bandwidths in range and azimuth, respectively, given by

$$\begin{cases} B_r = \gamma T_p \\ B_a = \dfrac{2Vc\,\varphi_{wa}}{f_c} \end{cases} \tag{45}$$

here γ denotes the chirp rate, T_p is the pulse duration, V represents the platform velocity, c is the speed of light, φ_{wa} denotes the antenna azimuth beamwidth, and f_c is the center frequency of the transmitted signal. The coefficient K_c is determined by the following expression:

$$\sigma_C = \mathrm{E}\left[\left|C(\tau,\eta)\right|^2\right] = \int_{-\frac{B_a}{2}}^{\frac{B_a}{2}} \int_{-\frac{B_r}{2}}^{\frac{B_r}{2}} S_C(u,v)\,\mathrm{d}u\mathrm{d}v \tag{46}$$

where σ_C denotes the average backscattering coefficient of clutter, which is set to −20 dB in the simulation.

The PSD of $N(\eta,\tau)$ is specified as

$$S_N(u,v) = K_n \begin{cases} -\dfrac{B_r}{2} \le u \le \dfrac{B_r}{2} \\ [5pt] -\dfrac{B_a}{2} \le v \le \dfrac{B_a}{2}. \end{cases} \tag{47}$$

K_n is determined by the following expression:

$$kB_rT_s = \mathrm{E}\left[\left|N(\tau,\eta)\right|^2\right] = \int_{-\frac{B_a}{2}}^{\frac{B_a}{2}} \int_{-\frac{B_r}{2}}^{\frac{B_r}{2}} S_N(u,v)\,\mathrm{d}u\mathrm{d}v \tag{48}$$

where k denotes the Boltzmann's constant and T_s represents the receiver noise temperature.

Given the spaceborne SAR parameters and the prior statistical information of the random signals $T(\tau,\eta)$, $C(\tau,\eta)$, and $N(\tau,\eta)$, JPDF of the $2N_d$-dimensional SAR image sample vectors $\mathbf{i}_0$ and $\mathbf{i}_1$ can be derived. To avoid unnecessary information loss, all available samples within the detection window are uniformly selected to construct the image sample vector, and accordingly, the value of N_d is given by

$$N_d = \left\lfloor \frac{c\,\tau_{\max}}{2\,\mathrm{res}_r} \right\rfloor \times \left\lfloor \frac{V_p\,a_{\max}}{\mathrm{res}_a} \right\rfloor. \tag{49}$$

Based on the target signal duration time, the detection window is set to $\tau_{max} \times \eta_{max}$. res_r and res_a are SAR range and azimuth resolutions. According to the construction of $\mathbf{i}_0$ and $\mathbf{i}_1$, the image sample vectors can be reshaped back into complex-valued matrix images $I_0(\tau,\eta)$ and $I_1(\tau,\eta)$ for visualization. Fig. 5 illustrates sample SAR images generated from random vectors drawn from the JPDF of $\mathbf{i}_0$ and $\mathbf{i}_1$, as described in (18) and (23), respectively. Fig. 5 illustrates the magnitude images of the complex-valued SAR image under target-present and target-absent conditions. It can be observed that the clutter within the detection window exhibits local texture structures, which interfere with extended target detection. The variability of the target scattering characteristics also affects the extended target detection performance.

Once the JPDF of $\mathbf{i}_0$ and $\mathbf{i}_1$ is obtained, UB-ETDA of the SAR system can be numerically computed according to the method in Section III, by setting $N_{sample} = 10^4$.

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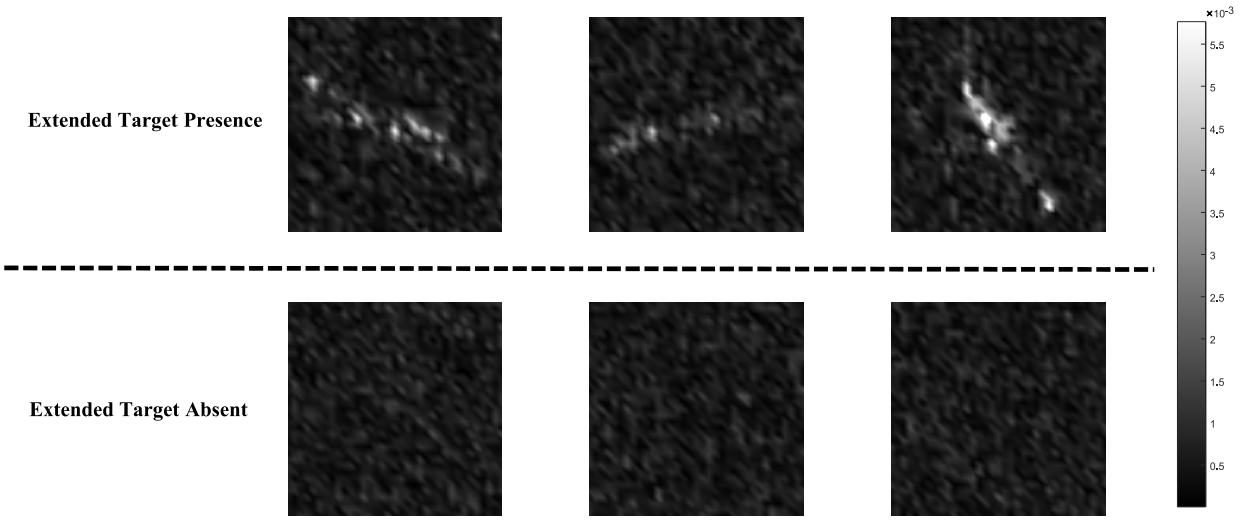


Fig. 5. Sample magnitude images of complex-valued SAR images under target present and target absent conditions.

The theoretically computed UB-ETDA is then compared with the ETDA achieved by extended-target detection methods. Three methods are selected for comparison to validate the correctness of UB-ETDA, including the generalized likelihood ratio test (GLRT) [42], the template matching method [43], and the subspace projection method [44]. Furthermore, to examine the achievability of UB-ETDA, an idealized Bayesian detection method is introduced, which assumes that the SAR detector has complete knowledge of the statistical distributions of SAR images under both target-present and target-absent hypotheses. Under this ideal assumption, the following Bayesian optimal detection rule is applied to the observed SAR image sample vector $\mathbf{i}$

$$\hat{V}_{\text{Bayes}} = \begin{cases} 1, & pf_1(\mathbf{i}) \geq qf_0(\mathbf{i}) \\ 0, & pf_1(\mathbf{i}) < qf_0(\mathbf{i}). \end{cases} \tag{50}$$

Since the target existence decision set adopted by the ideal Bayesian detector coincides with the optimal set $\mathcal{A}_*$ defined in (28), the ideal Bayesian detection method is theoretically capable of achieving the UB-ETDA.

Random samples following the $T(\tau,\eta)$, $C(\tau,\eta)$ and $N(\tau,\eta)$ statistical models are generated and processed by the SAR imaging system $h_{\text{Echo}}(\tau,\eta)$ and $h_{\text{Process}}(\tau,\eta)$ to produce SAR image samples, which are then tested using the three extended target detectors and an ideal detector described above; the detection results are collected to obtain the ETDA of each detector. The values of p and q are set to 0.5, and the SAR image sample number is set to 10^5.

Within the ranges specified in Table I, the chirp rate γ and azimuth beamwidth φ_{wa} are proportionally varied to achieve an SAR resolution ranging from 0.5 to 3 m. Meanwhile, the PRF f_r is set to 3000 Hz. Under different resolution settings, UB-ETDA is computed, while the ETDA of three detectors and the ideal Bayesian detector is estimated via Monte Carlo simulations. The corresponding results are shown in Fig. 6(a). The ETDA achieved by the three practical methods is lower than the UB-ETDA; it can also be observed that both the UB-ETDA and the ETDA of the detection methods first increase and then decrease as the resolution improves. This trend arises because an increase in resolution enlarges the system receive bandwidth, which enhances the representation of target structure; however, the clutter and noise energy received also grows with bandwidth, reducing the target-to-interference ratio and ultimately causing detection performance to decline after reaching a peak.

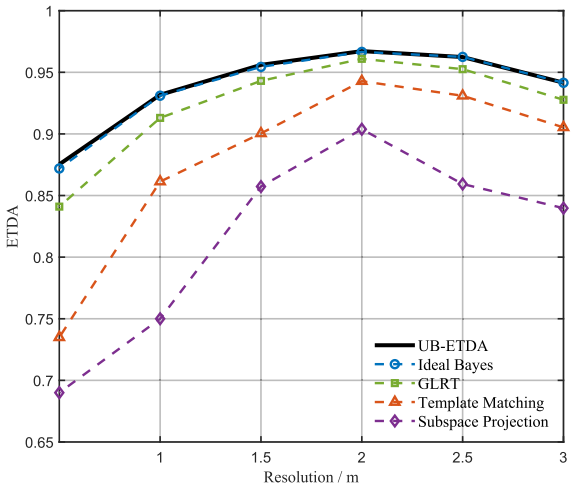
The PRF f_r is varied within the range specified in Table I. The azimuth beamwidth φ_{wa} is set at 0.0075 rad and the chirp rate γ at 4.185 MHz/μs, corresponding to a resolution of 2 m. As the PRF f_r varies, the UB-ETDA is computed, while the ETDA of the detection methods is evaluated using the same Monte Carlo procedure, with the results shown in Fig. 6(b). The ETDA achieved by the three practical methods is lower than the UB-ETDA, which verifies the correctness of the UB-ETDA. It can be observed that, as the PRF increases, both the UB-ETDA and the ETDA of the detection methods first increase and then remain approximately constant. This is because when the PRF is below the Nyquist sampling requirement determined by B_a, insufficient azimuth sampling leads to inadequate coherent accumulation of the received echoes. However, once the PRF exceeds this sampling requirement, further increasing the PRF has a negligible impact on the detection performance.

As shown in Fig. 6(a) and (b), the ETDA of the ideal Bayesian detection method reaches the UB-ETDA, and the ETDA of the GLRT detector is also close to the UB-ETDA. These results demonstrate the attainability of UB-ETDA as an upper bound on ETDA.

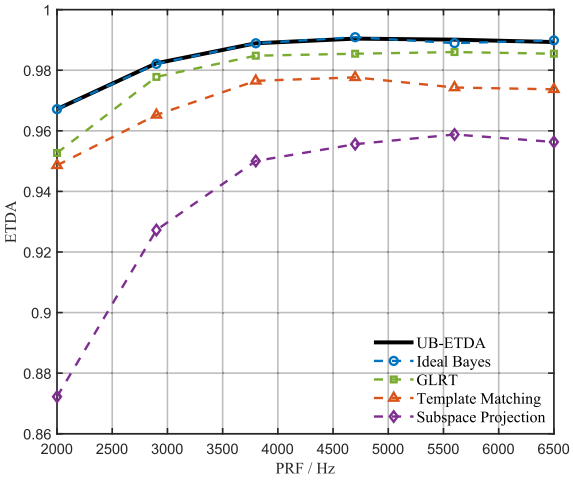
B. Spaceborne SAR Parameter Design Simulation

In this section, the proposed parameter design method is employed, where UB-ETDA is adopted as the objective function and the conventional performance metrics are imposed as constraints to obtain the optimal parameter set. In addition, two existing parameter design methods are considered for comparison. The extended target detection performance of the

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(a)



(b)

Fig. 6. Comparison of UB-ETDA and ETDA for different detectors under varying SAR parameters. (a) ETDA versus SAR resolution and (b) ETDA versus PRF.

TABLE II
FEASIBLE BOUNDARIES FOR PERFORMANCE METRICS

Performance Metrics	Value
$\text{res}_{\text{bound}}$	3 m
$\text{NESZ}_{\text{bound}}$	−19 dB
$\text{AASR}_{\text{bound}}$	−17 dB
$\text{RASR}_{\text{bound}}$	−17 dB
W_{bound}	20 km

three resulting parameter sets is then evaluated to validate the effectiveness of the proposed parameter design method.

In the simulation, the spaceborne SAR parameters and the statistical distribution of the target, clutter, and noise are set consistent with those in Section V-A. The parameters to be designed include the chirp rate γ, antenna azimuth beamwidth φ_{wa}, and PRF f_r, and their ranges are listed in Table I.

The value of feasible boundaries of traditional performance metrics in (43) is shown in Table II. The boundary values

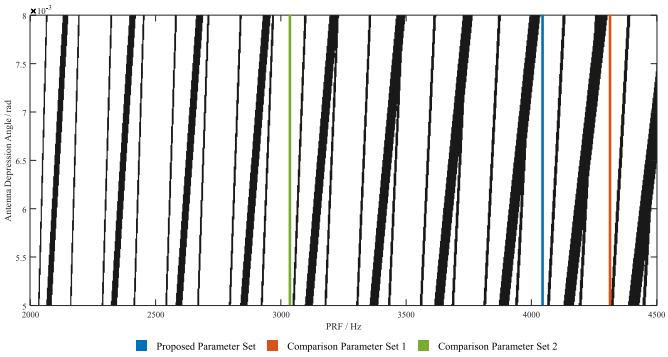


Fig. 7. Beam pointing of the proposed SAR parameter set on the zebra map.

of NESZ, ambiguity, and resolution are set to be consistent with those in the stripmap mode of TerraSAR-X [45]. The W_{bound} is set to 20 km. Since Wobs is determined solely by the antenna depression angle φ_d and φ_{wr}, which are predefined parameters, it is not affected by parameter optimization. Under the parameter settings listed in Table I, W_{obs} = 20.1km and satisfies the corresponding constraint. Therefore, this metric is not recalculated in the subsequent analysis.

The ε-DE algorithm is employed for parameter optimization [38], with its key parameters specified as follows: the population size is set to 20, the scaling factor is set to 0.7, the crossover rate is set to 0.9, the maximum generation is 500, and the control generation is set to 400. The proposed parameter set and the corresponding performance metrics are listed in the second column of Table III. The proposed beam pointing in the zebra map is illustrated by the blue solid line in Fig. 7. The black regions correspond to infeasible regions induced by the transmitted signal and nadir echo interference.

For comparison, parameter design method 1 adopts NESZ as the objective function, while imposing constraints on resolution, ambiguity, signal timing, and swath width [18], [46]. The values of predefined parameters and selectable ranges of the design parameters are consistent with those specified in Table I, while the boundaries of the performance metrics follow Table II. The resulting parameters and the corresponding performance metrics are listed in the third column of Table III, and the corresponding beam pointing in zebra map is shown as the orange solid line in Fig. 7.

Parameter design method 2 follows the approach in [20] with experimental configurations kept consistent with those in [20]. The prior information about target intensity required in the design process is obtained through statistical analysis of the EM simulation data used in this simulation. The predefined parameters are set identically to those in Table I. Since design method 2 does not impose constraints on parameter selectable ranges or performance metric boundaries, the constraints specified in Table I for parameter ranges and those in Table II for performance metrics are not applied in its implementation. However, the design method 2 typically yields poor AASR performance, the clutter backscattering coefficient σ_{C_2} is reformulated as follows:

$$\sigma_{C_2} = \sigma_C + \sigma_C \frac{\text{AASR}_{\text{lin}}}{1 - \text{AASR}_{\text{lin}}}. \tag{51}$$

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TABLE III
PROPOSED AND COMPARISON PARAMETER SETS WITH CORRESPONDING PERFORMANCE METRICS

Parameters / Performance Metrics	Proposed Set	Comparison Set 1	Comparison Set 2
PARAMETERS			
Chirp Rate γ	$4.19\,\mathrm{MHz}/\mu\mathrm{s}$	$2.79\,\mathrm{MHz}/\mu\mathrm{s}$	$10.49\,\mathrm{MHz}/\mu\mathrm{s}$
Azimuth Beamwidth φ_{wa}	$0.0075\,\mathrm{rad}$	$0.005\,\mathrm{rad}$	$0.0188\,\mathrm{rad}$
PRF f_r	$4045.6\,\mathrm{Hz}$	$4315.3\,\mathrm{Hz}$	$3035.0\,\mathrm{Hz}$
PERFORMANCE METRICS			
UB-ETDA	**99.0%**	94.1%	89.2%
Resolution	$2.0\,\mathrm{m}$	$3.0\,\mathrm{m}$	**0.8 m**
NESZ	$-19.4\,\mathrm{dB}$	**-24.8 dB**	$-11.2\,\mathrm{dB}$
AASR	$-25.1\,\mathrm{dB}$	$-24.6\,\mathrm{dB}$	$-2.4\,\mathrm{dB}$
RASR	$-27.2\,\mathrm{dB}$	$-26.0\,\mathrm{dB}$	$-34.6\,\mathrm{dB}$

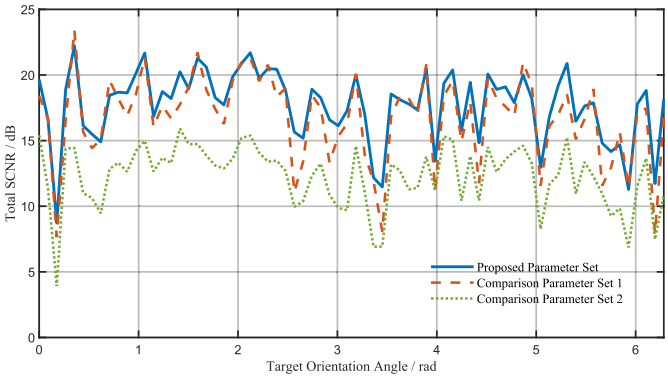


Fig. 8. Total-SCNR versus target orientation angle for different SAR parameter sets.

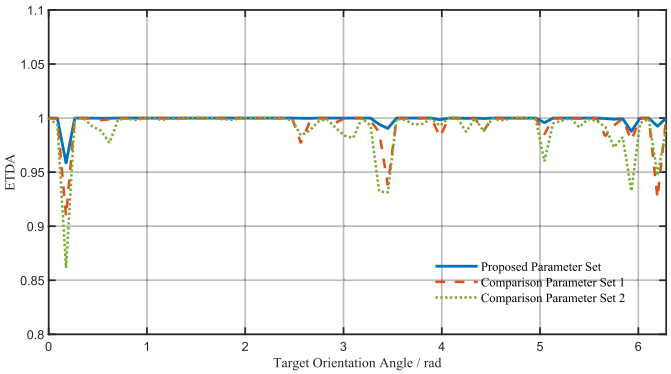


Fig. 9. ETDA versus target orientation angle for different SAR parameter sets.

The resulting parameters and the corresponding performance metrics are listed in the fourth column of Table III, and the corresponding beam pointing in zebra map is shown as the green solid line in Fig. 7.

Comparing the proposed parameter set and the comparison parameter set 1, it shows that although set 1 achieves better NESZ, its lower resolution results in insufficient representation of target structure. Consequently, its UB-ETDA is inferior to that of the proposed set. Comparing the proposed parameters with comparison parameter set 2, it can be observed that although set 2 offers higher resolution, its NESZ and AASR deteriorates significantly, limiting the signal-to-noise and signal-to-clutter characteristics. As a result, its UB-ETDA also remains lower than that of the proposed set. Given the definitions of resolution and NESZ, a negative correlation exists between them. The proposed parameter set achieves the best balance among resolution, target-to-noise ratio, and ambiguity suppression, thereby yielding the highest UB-ETDA.

The ratio of the energy of all target component pixels within the detection window to the energy of noise and clutter components in the same region is defined as the total signal-to-clutter-and-noise ratio (total-SCNR). Fig. 8 shows the total-SCNR as a function of target orientation angle for the proposed parameter set and the two comparison parameter sets. The results demonstrate that the proposed parameter set achieves favorable overall total-SCNR performance compared with the two comparison parameter sets. Compared with comparison parameter set 1, although the proposed parameter set exhibits slightly inferior performance at certain high total-SCNR orientation angles, it maintains stable and competitive performance under low total-SCNR conditions. Comparison parameter set 2, being more susceptible to strong clutter and noise interference, consistently yields lower total-SCNR across different target orientation angles. These results verify that the proposed parameter configuration can stably and effectively preserve extended target information while suppressing clutter and noise.

Under a given SAR parameter set, the statistical distributions of the target, background, and noise components in the SAR images can be derived, which allows the determination of the SAR image statistics distribution for a specific target orientation angle $\hat{\theta}_A$. By applying the ideal Bayesian target detection method (50) at each target orientation angle, the ETDA as a function of target orientation angle is obtained. ETDA versus target orientation for the proposed parameter set and the two comparative sets is shown in Fig. 9. The results indicate that the proposed parameters achieve consistently higher and stable ETDA across all orientations, particularly under challenging target orientation angles where detection performance is generally poor. Overall, the proposed parameter design provides stable and improved performance for extended target detection.

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VI. Conclusion

This article proposes a UB-ETDA based spaceborne SAR parameter design method to enhance the detection performance for extended targets. First, a signal model for SAR extended target detection is established, in which the statistical characteristics of the target, clutter, and noise components in SAR images are systematically characterized. Subsequently, based on the proposed signal model, the mathematical expression of UB-ETDA is rigorously derived, and a stable and efficient numerical computation method is developed. Then, a constrained optimization problem for spaceborne SAR parameter design is formulated, where UB-ETDA is adopted as the objective function and existing performance metrics are imposed as constraints. The proposed parameter set of the constrained optimization problem is obtained using an ε-DE algorithm. Finally, simulation results validate the correctness and attainability of UB-ETDA and demonstrate that the proposed spaceborne SAR parameter design method can significantly improve extended target detection performance.

By appropriately adapting the system model, the proposed UB-ETDA can be extended to other SAR modalities, such as airborne and bistatic SAR, for extended target detection performance evaluation. Furthermore, as the backscattering characteristics of a target under different orientation angles can represent its signature, the proposed extended target signal model has the potential to be extended to higher-level tasks such as target classification.

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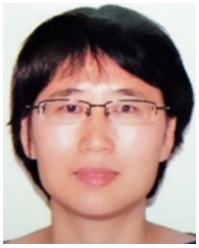
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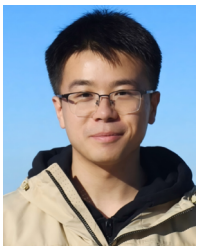
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